

## VIDEO PIPELINE OPTIMIZATION AND SCALING (MILESTONE 2)

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**GITHUB REPOSITORY:** <https://github.com/bhanuprakashvangala/VideoPipeline>

This report documents the Milestone 2 extension of our video object detection and tracking pipeline. Milestone 2 focuses on optimizing and scaling the detection stage by testing multiple model variants and resource allocations, then selecting an “optimal” configuration that satisfies a service-level agreement while maintaining tracking quality. We evaluated 19 configurations across YOLOv8 variants and scaling levels, measuring latency, throughput, accuracy, and cost.

### 1. PIPELINE OVERVIEW AND OBJECTIVE

The primary objective in Milestone 2 is to optimize and scale the Video Pipeline to meet performance SLAs while maintaining accuracy. Building on Milestone 1’s end-to-end detector + tracker prototype, we containerize model variants, introduce a service boundary (gRPC-ready), and evaluate horizontal scaling trade-offs (replicas  $\times$  CPU cores) under a target SLA.

The Milestone 2 pipeline follows a staged architecture:

#### Stage 1: Object Detection (variant + scale under test)

- Runs the selected detector variant and outputs bounding boxes with confidence scores.
- Applies confidence filtering (threshold  $\geq 0.3$ ) before tracking.

#### Stage 2: Object Tracking

- Associates detections across frames using OC-SORT to produce track IDs and trajectories.

#### Stage 3: Output + Logging

- Writes per-frame detections/tracks and summary metrics; logs latency/throughput/cost for analysis.

### 2. DATASET AND MODELS

Dataset: The MOT17-04-DPM dataset from the Multi-Object Tracking benchmark is

used. It contains 1,050 sequential frames of an urban street scene at 1920x1080 resolution, providing challenging conditions with occlusion and lighting variation.

- Total frames: 1,050
- Resolution: 1920×1080 pixels
- Content: Urban street scene with multiple pedestrians
- Format: Sequential JPEG images representing video frames
- Target class: Person detection only

Detection Models Implemented (Milestone 2):

- YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l (containerized variants)

Tracking Algorithm:

- OC-SORT (Observation-Centric SORT)
- Detection threshold: 0.3
- IoU threshold: 0.3
- Distance threshold: 100 pixels

### **Assertions and Metrics**

- Distance assertion (used as tracking accuracy): for each track\_id, the bbox-center displacement between consecutive frames must be  $\leq 100$  pixels.
- Tracking accuracy = (# valid transitions) / (# total transitions), where a transition is a consecutive (frame t, frame t+1) pair within the same track.

**Metrics reported:**

- Latency per frame (ms; compared to 500 ms SLA)
- Throughput (fps; compared to 20 fps SLA)
- Tracking accuracy (distance-assertion pass rate)
- Cost = (# replicas) x (CPU cores per replica)
- SLA Met (Latency  $\leq 500$  ms AND Throughput  $\geq 20$  fps)

### **3. PIPELINE CONFIGURATIONS**

Milestone 2 compares detector variants and scaling settings to understand cost–performance trade-offs under an SLA.

Configuration dimensions:

- Model variant: YOLOv8n / YOLOv8s / YOLOv8m / YOLOv8l
- CPU core allocations per replica: 1, 2, 4, 8
- Replica counts: 1–5 (tested across 19 total configurations)

- Hardware: CPU-only baseline vs multi-GPU runs

Horizontal scaling is simulated by dividing the workload across multiple “replicas” and processing batches in parallel, then measuring latency and throughput.

- Total test configurations: 19
- This ensures we test at least 4 model variants with 4 different core allocations
- Configuration format: (Model Variant, Replicas, CPU Cores per Replica)
  1. YOLOv8n with 1 replicas  $\times$  1 cores = 1 total cost
  2. YOLOv8n with 2 replicas  $\times$  1 cores = 2 total cost
  3. YOLOv8n with 2 replicas  $\times$  2 cores = 4 total cost
  4. YOLOv8n with 4 replicas  $\times$  1 cores = 4 total cost
  5. YOLOv8n with 4 replicas  $\times$  2 cores = 8 total cost

Figure 1. Milestone 2 methodology and configuration search space (variants, cores, replicas).

#### 4. PERFORMANCE RESULTS

The table below summarizes the key Milestone 2 configurations and the chosen “best” options under the target SLA (500 ms latency, 20 fps).

| 1  | Variant | Scale (Replicas) | CPU Cores | Cost | Latency (ms) | Throughput (fps) | Accuracy | Objects | SLA Met |
|----|---------|------------------|-----------|------|--------------|------------------|----------|---------|---------|
| 2  | YOLOv8n | 1                | 1         | 1    | 159.03       | 6.29             | 1        | 2511    | FALSE   |
| 3  | YOLOv8n | 2                | 1         | 2    | 134.31       | 7.45             | 1        | 2511    | FALSE   |
| 4  | YOLOv8n | 2                | 2         | 4    | 153.76       | 6.5              | 1        | 2511    | FALSE   |
| 5  | YOLOv8n | 4                | 1         | 4    | 133.3        | 7.5              | 1        | 2511    | FALSE   |
| 6  | YOLOv8n | 4                | 2         | 8    | 135.75       | 7.37             | 1        | 2511    | FALSE   |
| 7  | YOLOv8s | 1                | 1         | 1    | 222.15       | 4.5              | 1        | 2860    | FALSE   |
| 8  | YOLOv8s | 2                | 2         | 4    | 285.86       | 3.5              | 1        | 2860    | FALSE   |
| 9  | YOLOv8s | 3                | 2         | 6    | 228.17       | 4.38             | 1        | 2860    | FALSE   |
| 10 | YOLOv8s | 4                | 2         | 8    | 204.51       | 4.89             | 1        | 2860    | FALSE   |
| 11 | YOLOv8s | 4                | 4         | 16   | 232.42       | 4.3              | 1        | 2860    | FALSE   |
| 12 | YOLOv8m | 2                | 2         | 4    | 247.12       | 4.05             | 1        | 1310    | FALSE   |
| 13 | YOLOv8m | 3                | 2         | 6    | 485.52       | 2.06             | 1        | 3267    | FALSE   |
| 14 | YOLOv8m | 4                | 2         | 8    | 469.9        | 2.13             | 1        | 2714    | FALSE   |
| 15 | YOLOv8m | 4                | 4         | 16   | 514.18       | 1.94             | 1        | 2714    | FALSE   |
| 16 | YOLOv8m | 5                | 2         | 10   | 489.18       | 2.04             | 1        | 2714    | FALSE   |
| 17 | YOLOv8l | 2                | 4         | 8    | 405.78       | 2.46             | 1        | 870     | FALSE   |
| 18 | YOLOv8l | 3                | 4         | 12   | 834.7        | 1.2              | 1        | 1727    | FALSE   |
| 19 | YOLOv8l | 4                | 4         | 16   | 821.76       | 1.22             | 1        | 2111    | FALSE   |
| 20 | YOLOv8l | 5                | 4         | 20   | 869.2        | 1.15             | 1        | 1727    | FALSE   |

Figure 2. CPU-only results table (all configurations fail the throughput SLA).

| MILESTONE 2: PIPELINE PERFORMANCE TABLE |                  |           |      |              |                  |          |         |         |  |
|-----------------------------------------|------------------|-----------|------|--------------|------------------|----------|---------|---------|--|
| Variant                                 | Scale (Replicas) | CPU Cores | Cost | Latency (ms) | Throughput (fps) | Accuracy | Objects | SLA Met |  |
| YOLOv8n                                 | 1                | 1         | 1    | 701.78       | 1.42             | 1.0      | 2504    | False   |  |
| YOLOv8n                                 | 2                | 1         | 2    | 32.29        | 30.97            | 1.0      | 2504    | True    |  |
| YOLOv8n                                 | 2                | 2         | 4    | 26.47        | 37.78            | 1.0      | 2504    | True    |  |
| YOLOv8n                                 | 4                | 1         | 4    | 32.12        | 31.13            | 1.0      | 2504    | True    |  |
| YOLOv8n                                 | 4                | 2         | 8    | 25.94        | 38.55            | 1.0      | 2833    | True    |  |
| YOLOv8s                                 | 1                | 1         | 1    | 36.14        | 27.67            | 1.0      | 2860    | True    |  |
| YOLOv8s                                 | 2                | 2         | 4    | 25.65        | 38.98            | 1.0      | 2860    | True    |  |
| YOLOv8s                                 | 3                | 2         | 6    | 28.35        | 35.28            | 1.0      | 3368    | True    |  |
| YOLOv8s                                 | 4                | 2         | 8    | 27.54        | 36.31            | 1.0      | 2860    | True    |  |
| YOLOv8s                                 | 4                | 4         | 16   | 27.08        | 36.93            | 1.0      | 3663    | True    |  |
| YOLOv8m                                 | 2                | 2         | 4    | 29.21        | 34.23            | 1.0      | 3027    | True    |  |
| YOLOv8m                                 | 3                | 2         | 6    | 28.52        | 35.07            | 1.0      | 2714    | True    |  |
| YOLOv8m                                 | 4                | 2         | 8    | 26.35        | 37.96            | 1.0      | 2714    | True    |  |
| YOLOv8m                                 | 4                | 4         | 16   | 27.44        | 36.44            | 1.0      | 2714    | True    |  |
| YOLOv8m                                 | 5                | 2         | 10   | 28.34        | 35.29            | 1.0      | 2714    | True    |  |
| YOLOv8l                                 | 2                | 4         | 8    | 33.11        | 30.20            | 1.0      | 1727    | True    |  |
| YOLOv8l                                 | 3                | 4         | 12   | 29.40        | 34.01            | 1.0      | 1727    | True    |  |
| YOLOv8l                                 | 4                | 4         | 16   | 30.14        | 33.18            | 1.0      | 1727    | True    |  |
| YOLOv8l                                 | 5                | 4         | 20   | 31.21        | 32.04            | 1.0      | 2032    | True    |  |

Figure 3. GPU results table (runs on 4 GPUs; NVIDIA L4, 22GB; most configurations meet SLA).

### Analysis:

- CPU-only runs did not meet the throughput SLA ( $\geq 20$  fps); the fastest CPU configuration was  $\sim 7.5$  fps.
- GPU runs met SLA for most configurations (latency well under 500 ms and throughput  $> 20$  fps).
- Lowest cost meeting SLA: YOLOv8s, 1 replica x 1 core (cost 1), latency 36.14 ms, throughput 27.67 fps.
- Best throughput meeting SLA: YOLOv8s, 2 replicas x 2 cores (cost 4), latency 25.65 ms, throughput 38.98 fps.
- Best accuracy meeting SLA (distance-based metric): YOLOv8n, 2 replicas x 1 core (cost 2), latency 32.29 ms, throughput 30.97 fps.
- Per-step breakdown shows detection is the bottleneck ( $\sim 97\%$  of total time on average), so optimization should focus on detection (model choice, batching, and scaling).

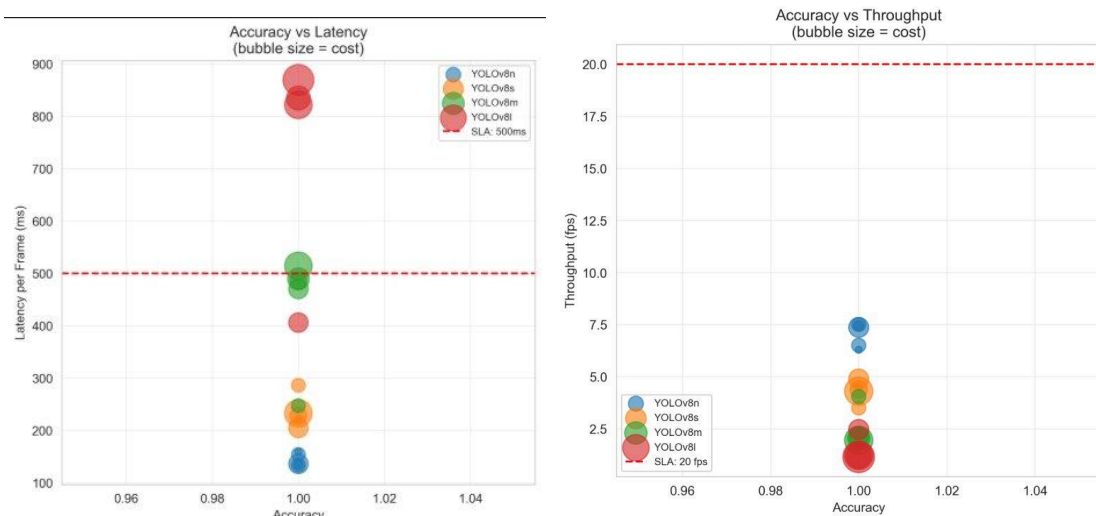


Figure 4a. CPU-only bubble plots: accuracy vs latency and accuracy vs throughput (bubble size = cost).

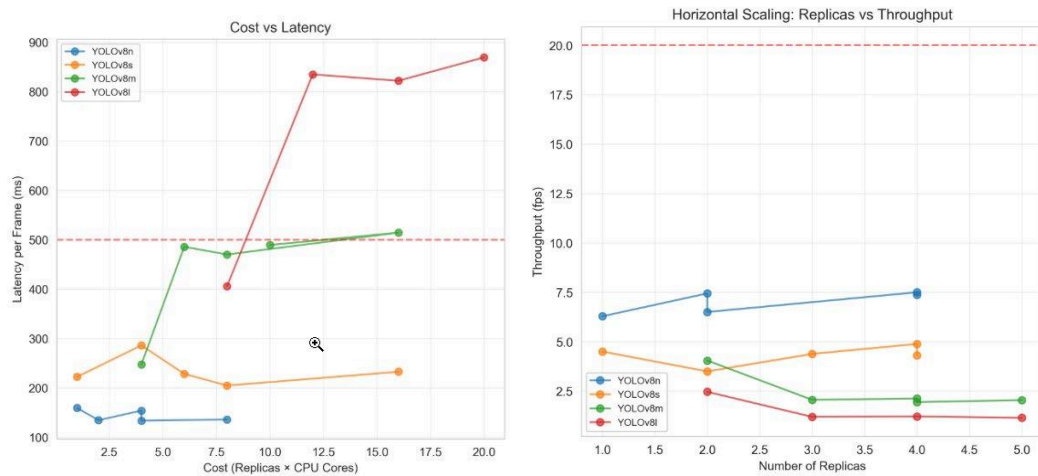


Figure 4b. CPU-only scaling trends: cost vs latency and replicas vs throughput.

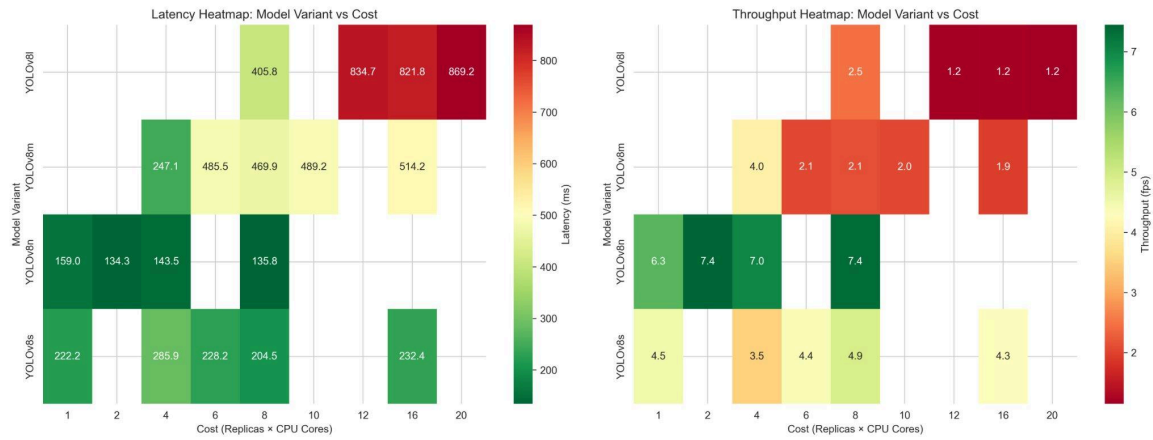


Figure 4c. CPU-only heatmaps: latency and throughput across variant vs cost.

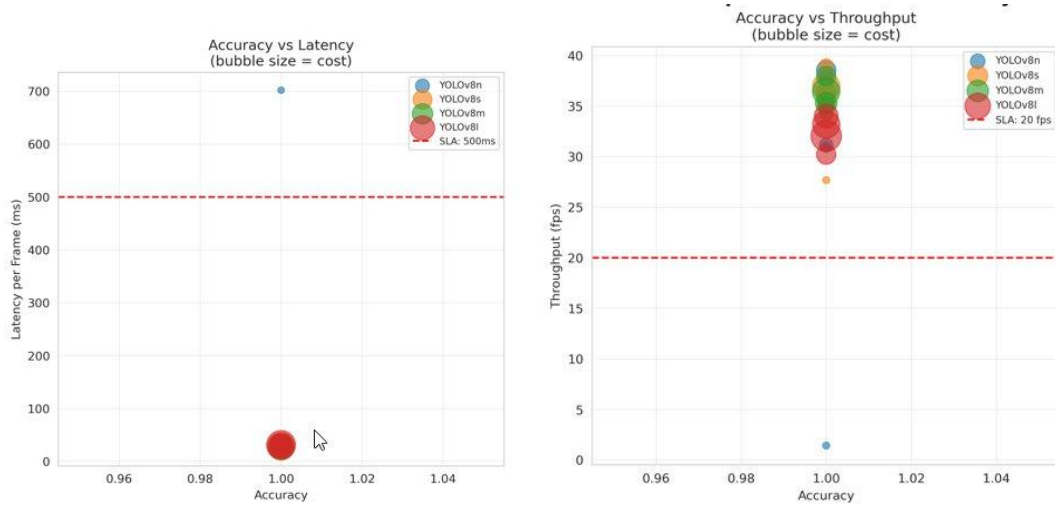


Figure 4d. GPU bubble plots: accuracy vs latency and accuracy vs throughput (bubble size = cost).

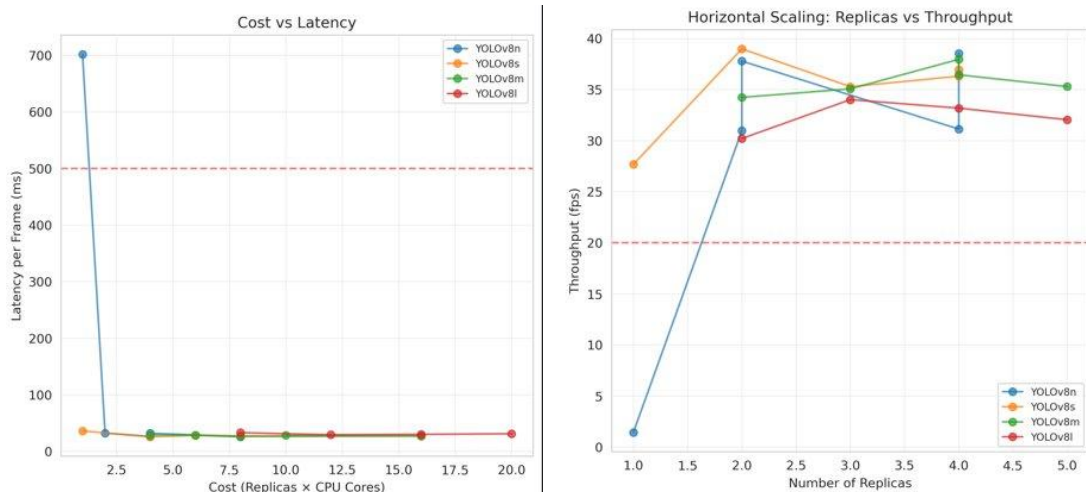


Figure 4e. GPU scaling trends: cost vs latency and replicas vs throughput.

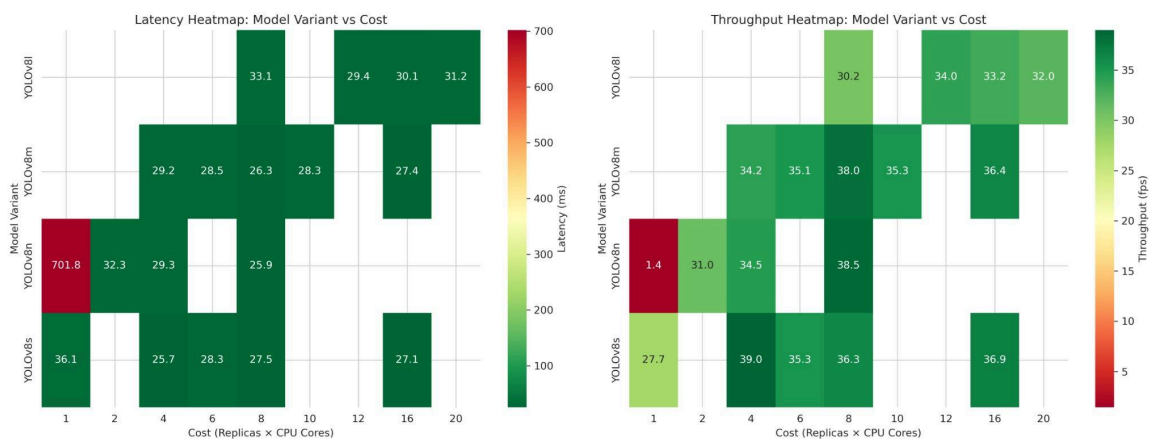


Figure 4f. GPU heatmaps: latency and throughput across variant vs cost.



| PER-STEP LATENCY BREAKDOWN |       |      |                    |                   |           |             |            |  |
|----------------------------|-------|------|--------------------|-------------------|-----------|-------------|------------|--|
| Variant                    | Scale | Cost | Detection Step (s) | Tracking Step (s) | Total (s) | Detection % | Tracking % |  |
| YOLOv8n                    | 1     | 1    | 70.10              | 0.07              | 70.18     | 99.9        | 0.1        |  |
| YOLOv8n                    | 2     | 2    | 3.16               | 0.07              | 3.23      | 97.8        | 2.2        |  |
| YOLOv8n                    | 2     | 4    | 2.58               | 0.07              | 2.65      | 97.4        | 2.6        |  |
| YOLOv8n                    | 4     | 4    | 3.14               | 0.07              | 3.21      | 97.8        | 2.2        |  |
| YOLOv8n                    | 4     | 8    | 2.52               | 0.08              | 2.59      | 97.3        | 3.1        |  |
| YOLOv8s                    | 1     | 1    | 3.52               | 0.09              | 3.61      | 97.5        | 2.5        |  |
| YOLOv8s                    | 2     | 4    | 2.48               | 0.09              | 2.57      | 96.5        | 3.5        |  |
| YOLOv8s                    | 3     | 6    | 2.73               | 0.10              | 2.83      | 96.5        | 3.5        |  |
| YOLOv8s                    | 4     | 8    | 2.46               | 0.29              | 2.75      | 89.5        | 10.5       |  |
| YOLOv8s                    | 4     | 16   | 2.60               | 0.11              | 2.71      | 95.9        | 4.1        |  |
| YOLOv8m                    | 2     | 4    | 2.83               | 0.09              | 2.92      | 96.9        | 3.1        |  |
| YOLOv8m                    | 3     | 6    | 2.77               | 0.08              | 2.85      | 97.2        | 2.8        |  |
| YOLOv8m                    | 4     | 8    | 2.56               | 0.08              | 2.63      | 97.3        | 3.0        |  |
| YOLOv8m                    | 4     | 16   | 2.67               | 0.08              | 2.74      | 97.4        | 2.9        |  |
| YOLOv8m                    | 5     | 10   | 2.76               | 0.08              | 2.83      | 97.5        | 2.8        |  |
| YOLOv8l                    | 2     | 8    | 3.27               | 0.04              | 3.31      | 98.8        | 1.2        |  |
| YOLOv8l                    | 3     | 12   | 2.90               | 0.04              | 2.94      | 98.6        | 1.4        |  |
| YOLOv8l                    | 4     | 16   | 2.98               | 0.04              | 3.01      | 99.0        | 1.3        |  |
| YOLOv8l                    | 5     | 20   | 3.07               | 0.05              | 3.12      | 98.4        | 1.6        |  |

BOTTLENECK ANALYSIS:  
Average Detection Step: 97.2% of total time  
Average Tracking Step: 2.9% of total time

→ DETECTION is the primary bottleneck (consuming 97.2% of pipeline time)  
Recommendation: Optimize detection step through model selection and horizontal scaling

Figure 5. Per-step latency breakdown (detection dominates total latency).

|                                                                                                                                                                       |                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. LOWEST COST (meeting SLA):</b><br>Variant: YOLOv8s<br>Scale: 1 replicas × 1 cores<br>Cost: 1<br>Latency: 36.14ms<br>Throughput: 27.67 fps<br>Accuracy: 1.0000   | <b>3. BEST THROUGHPUT (meeting SLA):</b><br>Variant: YOLOv8s<br>Scale: 2 replicas × 2 cores<br>Cost: 4<br>Latency: 25.65ms<br>Throughput: 38.98 fps<br>Accuracy: 1.0000                                                   |
| <b>2. BEST ACCURACY (meeting SLA):</b><br>Variant: YOLOv8n<br>Scale: 2 replicas × 1 cores<br>Cost: 2<br>Latency: 32.29ms<br>Throughput: 30.97 fps<br>Accuracy: 1.0000 | <b>4. BEST BALANCED (40% cost, 30% accuracy, 30% throughput):</b><br>Variant: YOLOv8s<br>Scale: 2 replicas × 2 cores<br>Cost: 4<br>Latency: 25.65ms<br>Throughput: 38.98 fps<br>Accuracy: 1.0000<br>Balanced Score: 0.637 |

Figure 6. Best configuration recommendations under SLA and weighted scoring.

## 5. IMPLEMENTATION

Milestone 2 implementation emphasizes reproducibility and deployability.



- Containerization: model dependencies are pinned (PyTorch + CUDA, ultralytics, ocsort, etc.).
- Orchestration: docker-compose.yml runs the pipeline environment and exposes JupyterLab for running experiments.
- Experiments: milestone2\_pipeline\_optimization.ipynb sweeps configurations and generates tables/plots for latency/throughput/cost.
- Tracking metric: calculate\_tracking\_accuracy() computes the distance-assertion pass rate ( $\leq 100$  px bbox-center movement between consecutive frames within the same track\_id).
- Service boundaries: the design supports model-to-model communication via gRPC to enable independently scalable detection and tracking services.

## 6. TEAM ORGANIZATION AND COLLABORATION

Work was split to parallelize development and integration:

- Bhanu Prakash Vangala: repository integration, experiment documentation, and results presentation.
- Nolan Rink: YOLOv8 variant/scaling experiments, metric collection, and analysis write-up.

Collaboration occurred via GitHub for code sharing, Microsoft Teams for real-time communication, and Google Drive for document sharing.

## 7. DELIVERABLES

Milestone 2 Deliverables:

- **Notebook:** milestone2\_pipeline\_optimization.ipynb (variants, scaling experiments, plots)
- **Plot comparing model version, accuracy and latency, and another plot comparing model version, accuracy and throughput:** Figures 4a - 4f
- **Table which reports Variant, Scale, Latency, Cost, Accuracy across the different steps of your pipeline:** Figure 5
- **Report your analysis: which model version will you choose and at how many resources:** Figure 6
- **Report:** This document
- **Project Presentation:** In class

## 8. Docker Environment

Milestone 2 was executed in a Docker-based environment to ensure reproducibility.

- docker-compose.yml builds/runs the environment, maps port 8888 for JupyterLab, and mounts ./notebooks and ./results for persistence.
- GPU benchmarking was performed on CUDA using NVIDIA L4 GPUs (22 GB VRAM)

each); the reported GPU table summarizes runs using 4 GPUs.

- Typical use: `docker compose up --build`, then run `milestone2_pipeline_optimization.ipynb` inside JupyterLab.

```
Using device: cuda
GPU Name: NVIDIA L4
GPU Memory: 22.0 GB
```

*Figure 7. GPU hardware used for benchmarking (CUDA device: NVIDIA L4, 22 GB).*

## 9. CONCLUSIONS

Milestone 2 extends the Milestone 1 pipeline by introducing optimized, scalable detector variants and a repeatable benchmarking methodology under an SLA. Results show that CPU-only execution cannot meet the required throughput target, while GPU-backed configurations can exceed the SLA with low latency. Across the tested space, YOLOv8s at 2 replicas  $\times$  2 cores provided the best throughput and best balanced score at moderate cost, making it the recommended default. Per-step latency analysis indicates the detector is the dominant bottleneck, so future optimizations should focus on detector efficiency, batching, and scaling policies.